**CONFIDENTIAL** **MOCK EXAM 2026**

# GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 6 printed pages*

**PART A (80 MARKS)**

**QUESTION 1**

a) List TWO (2) differences between morals and laws as described in the lecture on ethics and computer security.

(2 marks)

b) Discuss how a professional code of ethics and conduct guides a computing professional in carrying out security-related work. Support your discussion with six (6) distinct obligations drawn from the ACM code.

(6 marks)

**QUESTION 2**

a) Explain how a substitution cipher works, stating what serves as the key and which of Shannon's principles it corresponds to.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

(6 marks)

**Vigenere tableau (A = 0)**

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| B | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
| C | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
| D | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
| E | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
| F | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
| G | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
| H | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
| I | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
| J | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
| K | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
| L | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
| M | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
| N | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
| O | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
| P | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
| Q | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| R | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
| S | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
| T | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
| U | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| V | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| W | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
| X | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
| Y | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
| Z | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |

## QUESTION 3

a) Explain the Chinese Wall model as a formal security model, stating the concern it addresses and how its access decisions differ from static policies.

(2 marks)

b) Describe the access control matrix, the access control list and the capability list as access control structures, giving for each structure one advantage and one disadvantage, and state two requirements that any access control structure must satisfy.

(6 marks)

## QUESTION 4

a) Explain two characteristics of Trojan horse malware, including how it disguises itself and what harm it can cause once active.

(2 marks)

b) State six techniques or behaviours by which malware propagates itself or conceals its activity on an infected system, drawing on the historical and technical examples covered in the course.

(6 marks)

**QUESTION 5**

a) State TWO (2) typical locations where network-based IDS sensors should be placed to protect an organization's most valuable assets.

(2 marks)

b) Refer to your answer in (a). Based on the scenario in (a), explain how the organization can strengthen its preventive network security mechanisms by describing the role of a firewall, a proxy server, and network segmentation in controlling and protecting network traffic.

(6 marks)

**QUESTION 6**

a) Explain two ways in which a documented code of ethics shapes the ethical conduct of computing professionals in security-related work.

(2 marks)

b) Explain six responsibilities that a computing professional owes to users and the organisation when handling private data in security-related work, drawing on the lecture's discussion of ethics and professional conduct.

(6 marks)

**QUESTION 7**

a) List two classes of threats described in the course and, for each, state the security objective that is affected and the type of harm it represents.

(2 marks)

b) Describe the Samy worm as a case study of a cross-site scripting attack. In your answer, explain how the malicious script was injected, how it spread between users, how the victim's browser was made to act, and the scale of the resulting infection.

(6 marks)

**QUESTION 8**

a) State the two stages of the biometric authentication process and, for each stage, name the activities it involves.

(2 marks)

b) Explain the categories of biometric applications and the characteristics of the biometric technologies used for authentication, giving examples of each category and the practical issues associated with these technologies.

(6 marks)

**QUESTION 9**

a) State two ways in which an intrusion detection system contributes to detecting and defending against attacks on a system.

(2 marks)

b) State six points that distinguish the three security concepts of vulnerability, threat and attack, and give one countermeasure example for a threat class, drawing on the course material.

(6 marks)

**QUESTION 10**

a) Explain how a buffer overflow arises in a program such as the one using strcpy() into a fixed-size local buffer, and state what an attacker can achieve by exploiting it.

(2 marks)

b) Windows is often described as offering rich but complicated protection facilities. Explain six aspects of Windows security mechanisms, covering how subjects and their credentials are handled, how resources and policies are organised, and why the complexity of these mechanisms can itself become a security problem.

(6 marks)

**PART B (20 MARKS)**

**QUESTION 1**

A small company runs a shared Linux server where staff log in with user names and passwords. After several accounts were guessed by an attacker, the administrator is asked to strengthen how passwords are protected and managed on the system.

a) Explain five password security controls that the administrator can apply to protect the accounts on this server, and state how passwords should be stored.

(5 marks)

b) Discuss how single sign-on and Kerberos could reduce the password burden on the staff of the company in the scenario, and explain the design criteria that make Kerberos suitable for authenticating users across its distributed servers.

(5 marks)

**QUESTION 2**

A hospital wants to prove to auditors that every server booted only approved software. The vendor proposes placing a tamper-resistant chip on each motherboard and building a chain of trust from that chip upward through the boot software.

a) Referring to the hospital's proposed tamper-resistant chip, explain how a chain of trust is established from the hardware up to the running platform, and state what the chip is responsible for.

(5 marks)

b) The vendor's tamper-resistant chip is said to support NGSCB. State five things about the Nexus and the Nexus Computing Agent that the hospital's security team should understand before relying on this design.

(5 marks)

**END OF QUESTION PAPER**